Learning Objective 2.5: I can explain the theory behind antenna pattern measurement, including anechoic chambers, near-field to far-field transformations, and standard gain horns.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Sizing the range.** A 0.75 m aperture antenna is to be pattern-tested at 8 GHz. Take $c = 3 \times 10^8$ m/s throughout.

   (a) Find the wavelength and the minimum far-field range length for this antenna.

   r_{ff} = 30.0 m

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{8 \times 10^9} = 0.0375 \text{ m}$$

$$r_{ff} = \frac{2D^2}{\lambda} = \frac{2\,(0.75)^2}{0.0375} = \frac{1.125}{0.0375} = 30.0 \text{ m}$$

   (b) The only chamber available puts the source 12 m from the positioner. Find the edge phase error, in degrees, that this range imposes across the antenna.

   $\Delta\phi_{max}$ = $56.3°$

$$\Delta\phi_{max} = \frac{\pi D^2}{4\lambda r} = \frac{\pi\,(0.75)^2}{4\,(0.0375)\,(12)} = \frac{1.767}{1.80} = 0.982 \text{ rad}$$
$$= 56.3°$$

   Equivalently, $r/r_{ff} = 12/30 = 0.40$, and the edge error scales as the inverse of that ratio: $22.5°/0.40 = 56.3°$.

   (c) A colleague must test a 2.4 m reflector at 35 GHz. Find the far-field range length her measurement would require.

   r_{ff} = 1.34 km

$$\lambda = \frac{3 \times 10^8}{35 \times 10^9} = 8.57 \times 10^{-3} \text{ m}$$

$$r_{ff} = \frac{2\,(2.4)^2}{8.57 \times 10^{-3}} = \frac{11.52}{8.57 \times 10^{-3}}$$
$$= 1344 \text{ m} \approx 1.34 \text{ km}$$

   (d) Your colleague has no 1.3 km of chamber and no budget for a 1.3 km outdoor range. Name the two measurement approaches that solve her problem, and state in one sentence what each one trades away to do it.

   A **compact range** puts a precision offset-parabolic reflector in the room and collimates the feed's spherical wave into a plane wave over a short distance; it trades away money and simplicity, since the usable quiet zone is only about 50–60% of the reflector aperture (so the reflector must be far larger than the antenna) and the rim must be serrated or rolled to keep edge diffraction out of the test volume.

   A **near-field scanner** measures amplitude and phase on a surface a few wavelengths from the aperture and transforms the result to the far field; it trades away speed and simplicity, since it needs a phase-coherent receiver, half-wavelength sampling over a large surface, and probe compensation.

2. **Reading a suspect pattern.** A short range does not fail obviously; it returns a pattern that still looks like a pattern.

   (a) In one or two sentences, explain physically why a source at finite range produces a phase error across the antenna under test that grows as the *square* of the distance from the aperture center.

   The incident wavefront is a sphere of radius r centered on the source, not a plane. A point a distance x off the aperture center is $\sqrt{r^2 + x^2} \approx r + x^2/2r$ away from the source, so the extra path – and therefore the extra phase lag – is $x^2/2r$. The linear term vanishes because the geometry is symmetric about boresight, leaving the quadratic term as the leading error.

   (b) An antenna with $D/\lambda = 25$ is measured at half its far-field distance. Find the edge phase error in degrees.

   $\Delta\phi_{max}$ = $\boxed{45^\circ}$

   The far-field criterion is exactly $\pi/8 = 22.5^\circ$ of edge error, and the error scales as $1/r$:

$$\Delta\phi_{max} = \frac{22.5^\circ}{r/r_{ff}} = \frac{22.5^\circ}{0.5} = 45^\circ$$

   Note that this does not depend on D/λ at all – only on how short the range is relative to $2D^2/\lambda$.

   (c) In that same measurement the first null reads -16 dB instead of going deep, but the measured half-power beamwidth agrees with prediction to better than 2%. Explain, in one or two sentences, how both statements can be true at once.

   Quadratic phase error is a defocus, and a defocus destroys the precise cancellation that creates a null long before it changes the width of the main beam. The nulls depend on complete destructive interference across the whole aperture, so a 45° edge error fills them to about -16 dB ; the half-power points sit where the pattern is still dominated by the aperture's size, which the phase error has not changed. This is exactly why $2D^2/\lambda$ is a main-beam and gain criterion, not a sidelobe criterion.

   (d) A customer needs the -25 dB sidelobe of this antenna verified. Using the fact that a range at $2D^2/\lambda$ fills the first null to about -22 dB , at $5D^2/\lambda$ to about -30 dB , and at $10D^2/\lambda$ to about -36 dB , recommend a range length and defend it in one sentence.

   Recommend $r \geqslant 10D^2/\lambda$, i.e. five times the usual far-field distance. At $2D^2/\lambda$ the range's own artifact sits at -22 dB , which is *above* the -25 dB feature being measured – the measurement would be reporting the range, not the antenna. At $5D^2/\lambda$ the artifact is -30 dB , only 5 dB below the sidelobe, which is still worth several dB of error. At $10D^2/\lambda$ the artifact is -36 dB , a comfortable 11 dB below the feature, and the sidelobe level can be quoted with confidence.

3. **Specifying the chamber.** An anechoic chamber is not a room with no reflections. It is a room with reflections below a stated level.

   (a) Absorber reflectivity is quoted at normal incidence and degrades at grazing incidence. Explain why, and state which chamber surfaces this makes the critical ones.

   The pyramids work as a gradual impedance taper: at normal incidence a wave travels a long way down into the lossy foam before it can reflect, so it is absorbed. At grazing incidence the wave sees the pyramid tips as a nearly continuous surface and travels only a short distance in the lossy material, so it reflects far more strongly. The critical surfaces are therefore the **side walls, ceiling, and floor** in the specular region between the source and the antenna under test, where the stray rays strike at a slant.

   (b) A single stray reflection arrives 40 dB below the main beam. Find the worst-case error, in dB, that it adds to a measurement of (i) the main beam peak and (ii) a -30 dB sidelobe.

   main beam = $\boxed{\pm 0.09 \text{ dB}}$

   sidelobe = $\boxed{+2.4/-3.3 \text{ dB}}$

   The stray adds in and out of phase with the wanted signal. Work with voltage ratios. For the main beam, the ratio is $10^{-40/20} = 0.0100$:

   $$20 \log_{10}(1 \pm 0.0100) = +0.09 \text{ dB} \ / \ -0.09 \text{ dB}$$

   For a -30 dB sidelobe the stray is only 10 dB below the wanted signal, so the ratio is $10^{-10/20} = 0.316$:

   $$20 \log_{10}(1 \pm 0.316) = +2.4 \text{ dB} \ / \ -3.3 \text{ dB}$$

   (c) You must verify a -35 dB sidelobe to within ± 1 dB. Find the chamber reflectivity, relative to the main beam, that this requires.

   reflectivity = $\boxed{\approx -55 \text{ dB}}$

   Let x be the stray-to-signal voltage ratio at the sidelobe. The negative-going error is the binding one:

   $$\begin{aligned} 20 \log_{10}(1 - x) &\geqslant -1 \text{ dB} \\ 1 - x &\geqslant 10^{-1/20} = 0.891 \\ x &\leqslant 0.109 \quad (-19.3 \text{ dB}) \end{aligned}$$

   The stray must therefore sit 19.3 dB below the sidelobe, which is 19.3 dB below -35 dB:

   $$-35 \text{ dB} - 19.3 \text{ dB} = -54.3 \text{ dB}$$

   Specify -55 dB over the band and quiet zone.

   (d) Your antenna under test is 1.5 m across. The chamber datasheet advertises a 1.2 m quiet zone at -45 dB. State the problem in one sentence, and say what happens to the measurement if you proceed anyway.

   The antenna does not fit inside the volume the chamber is certified for, so its outer 0.15 m on each side is illuminated by fields that carry no reflectivity guarantee at all. Proceeding means the aperture edges – the region that sets the sidelobe structure – are being measured in an ordinary room, and any sidelobe or null result from the run is indefensible even though the main beam and gain will still look fine.

4. **Two ways to an absolute number.** Pattern shape needs only a normalization; gain needs a calibration.

   (a) At 12 GHz a standard gain horn with a calibrated gain of 20.4 dBi receives -41.2 dBm. Swapping in the antenna under test, with nothing else touched, gives -29.8 dBm. Find the gain of the antenna under test.

   G_{AUT} = 31.8 dBi

$$\Delta P = P_{AUT} - P_{SGH} = -29.8 - (-41.2)$$
$$= +11.4 \text{ dB}$$
$$G_{AUT} = G_{SGH} + \Delta P = 20.4 + 11.4 = 31.8 \text{ dBi}$$

   (b) The transmit power, the source antenna's gain, the range length, and the cable losses were all unknown in part (a), yet the answer is exact. Explain in one or two sentences why they do not matter.

   Write Friis for each of the two measurements. Every one of those quantities appears identically in both, because nothing but the antenna on the positioner changed, so all of them cancel in the difference $P_{AUT} - P_{SGH}$. The comparison method measures a *ratio* of gains, and the horn's calibration supplies the only absolute number in the room.

   (c) No standard is available, so you set up two identical antennas facing each other at $R = 15$ m and 10 GHz. With $P_t = +10$ dBm you measure $P_r = -30.0$ dBm. Find the gain of one antenna.

   G = 18.0 dBi

   Friis in dB, with $G_t = G_r = G$:

$$(P_r - P_t)_{dB} = 2G + 20\log_{10}\left(\frac{\lambda}{4\pi R}\right)$$

   With $\lambda = 0.03$ m:

$$\frac{\lambda}{4\pi R} = \frac{0.03}{4\pi\,(15)} = 1.592 \times 10^{-4}$$
$$20\log_{10}\left(1.592 \times 10^{-4}\right) = -76.0 \text{ dB}$$

$$2G = (-30.0 - 10.0) - (-76.0) = 36.0 \text{ dB}$$
$$G = 18.0 \text{ dBi}$$

   (d) Back in part (a): the standard gain horn is well matched, but the antenna under test has a VSWR of 2.0. Find the size of the resulting error, say which direction it pushes the answer, and state how you would report the result.

   VSWR $= 2.0$ gives $|\Gamma| = (2-1)\,/\,(2+1) = 1/3$, so the fraction of incident power accepted is $1 - |\Gamma|^2 = 1 - 1/9 = 0.889$:

$$\text{mismatch loss} = -10\log_{10}(0.889) = 0.51 \text{ dB}$$

   The antenna under test delivers 0.51 dB less power than a matched antenna of the same gain would, so the comparison method reads 0.51 dB **low**: 31.8 dBi measured against 32.3 dBi of true gain. Report the measured figure as *realized gain*, which is the system-level number, and quote the mismatch-corrected 32.3 dBi separately if the customer wants gain proper.

5. **Near-field scanning and the cut list.** The last two pieces of the midterm: how you measure when the range cannot exist, and what a complete characterization actually contains.

   (a) A near-field scanner must record both amplitude *and* phase at every sample point. Explain in one or two sentences why magnitude alone will not do.

   The near-field to far-field step is a Fourier transform, and a Fourier transform operates on the complex field. Magnitude alone does not determine the transform – infinitely many different phase distributions share the same magnitude on the scan plane and produce entirely different far-field patterns – so a scanner needs a phase-coherent receiver and phase-stable cabling.

   (b) Explain, in two or three sentences, why the near-field to far-field transform is exact rather than a curve fit. Name the earlier result it rests on.

   L6 established that the far field is the Fourier transform of the source distribution, with $k\sin(\theta)$ acting as the spatial frequency variable. Fourier transforms are invertible, so the complex tangential field sampled on a surface enclosing the antenna determines the field everywhere outside that surface – uniquely, with no fitting or assumption involved. The transform is the same relationship that produced the pattern in the first place, run in the other direction.

   (c) A planar scan at 18 GHz covers a 1.5 m by 1.5 m plane at the coarsest spacing the sampling rule allows. Find the required sample spacing and the total number of sample points.

   Δ = $\boxed{8.33\text{ mm}}$

   N = $\boxed{32{,}761}$

$$\lambda = \frac{3\times10^8}{18\times10^9} = 0.01667\text{ m}$$
$$\Delta = \frac{\lambda}{2} = 8.33\times10^{-3}\text{ m} = 8.33\text{ mm}$$

   Along one edge there are $1.5/8.33\times10^{-3} = 180$ intervals, hence 181 points:

$$N = (181)^2 = 32{,}761 \approx 3.3\times10^4\text{ points}$$

   At a few points per second this is several hours of scanning, which is the principal cost of a near-field range.

   (d) You are handed a linearly polarized pyramidal horn and told to characterize it. List the pattern cuts and polarization configurations that make up a minimum complete characterization, state how many sweeps that is, and name one thing that set of sweeps still will not tell you.

   Two principal-plane great-circle cuts, each run in two polarization configurations:

   1. E-plane, co-polarized (source aligned with the horn's E-field)
   2. E-plane, cross-polarized (source rotated 90°)
   3. H-plane, co-polarized
   4. H-plane, cross-polarized

   That is **four sweeps**. What it will not tell you is the pattern in the diagonal (45°) planes, which for a rectangular aperture is precisely where the highest sidelobes usually sit; a conical cut or a full spherical scan is needed for those.

Documentation: ____